

Stratified Randomization in Clinical Trials

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1 Introduction

Clinical trials are studies that test experimental treatments to determine if they are better, based on a chosen response variable, than current treatments. In a trial, participants are split into two groups: treatment and control. The group assignment process is performed randomly to increase the validity of the trial results. However, there can be covariates, variables that are not of interest but can impact the response variable, present within a clinical trial population. However, stratification can be implemented to account for covariates, in turn reducing variance and increasing the understanding of the efficacy of treatment.

2 Clinical Trials

2.1 What is a Clinical Trial?

A clinical trial is a study that tests an experimental treatment against the current treatment to determine if it is more effective. If a study proves efficacy, the public can then use the new treatment. The goal of a clinical trial is to create a new standard of care. Efficacy is quantified by taking the difference in mean outcomes between treatment and control groups. If the difference is considered statistically significant based on criteria set before the trial begins, then the trial is successful. A clinical trial determines whether a treatment is safe and effective before testing if it is better than the standard of care.

In a clinical trial, hypothesis testing is done. The null hypothesis, or H_0 , is that the difference in the actual and expected means is zero. The alternative hypothesis, or H_A , is that the difference in means is greater or less than zero. This depends on if the response variable should be increasing or decreasing. It is possible for a treatment to be worse than the current standard, but it is hoped for this to not be the case. In a clinical trial, H_0 : Current standard of care and H_A : The new treatment makes a difference.

2.2 Sample Clinical Trial: Moderna COVID-19 Vaccine

To solidify these concepts, they can be seen in a real-life example of the Moderna COVID-19 vaccine trial. In this case, there was no existing standard of care and a new vaccine had to be created to prevent COVID-19 infection. The response variable in the trial was whether a patient had symptomatic, laboratory-confirmed COVID-19 in roughly the first two months after full vaccination. The researchers randomly assigned placebo, which in this case was saline, or treatment, the vaccine. After the trial ended, researchers measured and compared the rates of positive COVID tests in both trial groups.

In this trial, H_0 : The efficacy of the vaccine is 30% or less and H_A : The efficacy of the vaccine is above 30%. The hypothesis was testing at a significance level, α , of 0.001. α is the chance of seeing results assuming that the null hypothesis, in this case the α level means that there would be a 0.1% chance of seeing the true results under the hypothesis that the efficacy of the vaccine is 30% or less. The results of the trial were that 94.1% efficacy was found in preventing COVID-19 infection through vaccination.

3 Random Assignment

In the case of simple randomization, each participant has a 50/50 chance of being assigned to the control or treatment group. However, this leads to the possibility of having uneven sizes of the control and treatment groups. In addition to the possibility of uneven groups, there could be covariates, such as age or gender, that may influence the results of the trial.

Stratification aims to avoid the influence of covariates. Strata is a subset of the population based on a characteristic. Going back to the Moderna clinical trial, one of the covariates researchers stratified by was age. In particular, the participants were divided into under 65 years old and 65 years or older. By splitting

participants by a characteristic, it can be determined if the results of the trial were caused by that characteristic or because of the treatment.

In order to evenly divide participants, after stratifying, into control and treatment groups, stratified block randomization can be done. The first step is to decide on a block size. A standard block size is 4. Then, the treatment assignment combinations within that block size are determined. For example, if A = treatment and B = control, the block assignment possibilities are AABB, ABAB, ABBA, BBAA, BABA, and BAAB. Lastly, each block is randomly assigned an assignment combination. This process promotes evenness of treatment and control assignments in groups.

4 Power

In a trial, it is ideal for power to be high. Power is the probability of avoiding a type II error which occurs when the null hypothesis is rejected when it should not be. Statistical power is related to sensitivity, which is the likelihood of a significance test detecting an effect where there is one. Therefore, when power is higher there is a larger chance of a test detecting a true effect of treatment. Power is influenced by sample size, effect size, and significance level. Power analysis consists of statistical power, sample size, significance level, and the effect size and if three values are known, the fourth can be calculated. Often times, it is used to determine the smallest sample size required for a desired power.

Stratification balances important baseline covariates across the treatment and control groups, which reduces unexplained outcome variation and makes treatment differences easier to detect. In turn, stratification increases power and higher power suggests higher reliability in trial results. Stratification increasing power is proven by Model-Robust Inference for Clinical Trials that Improve Precision by Stratified Randomization and Covariate Adjustment by Wang, B., Susukida, R., Mojtabai, R., Amin-Esmaeili, M., Rosenblum, M.

5 Conclusion

Stratification in clinical trials divides the population into subgroups to account for covariates and potential confounding variables, reduce variance, and increase power. All together, understanding of treatment efficacy is increased.

6 References

- Baden, Lindsey R., et al. "Efficacy and Safety of the mRNA-1273 SARS-CoV-2 Vaccine." *New England Journal of Medicine*, vol. 384, no. 5, 2021, pp. 403–416. <https://doi.org/10.1056/NEJMoa2035389>
- Wang, Bingkai, et al. Model-Robust Inference for Clinical Trials that Improve Precision by Stratified Randomization and Covariate Adjustment. 2020. arXiv, <https://arxiv.org/abs/1910.13954>